

huge demand for land may turn out to be a challenge unless the draconian land acquisition laws in the country are streamlined. Effective planning of projects also requires site-specific pre-feasibility analysis and assessment of local household and agricultural demand in the area.

Incentivising polysilicon manufacturing

The growth of solar industry in India opens a huge market for polysilicon. In the absence of domestic capacity, India is heavily dependent on imports. Polysilicon manufacturing involves high initial investment and also needs a large volume to achieve RoI. Local manufacturing needs special incentives to compete with global manufacturers. These incentives may either be in the form of tax sops or policies that stipulate special preference for local vendors. This, in turn, will benefit the solar ecosystem in terms of productivity, foreign trade and employability, among others. Adani and SunEdison have announced an investment of about US\$ 4 billion in Gujarat for a vertically-integrated polysilicon-refining facility.

Enabling efficient-energy storage

Availability and reliability are imperatives for any successful energy source. Energy storage systems (ESSes) and charging mechanisms play a critical role for ensuring reliability and efficiency of solar power sources in the country, especially during the season of low solar light and in the hinterlands of the country that are prone to frequent power blackouts. Moreover, an efficient ESS also helps to ensure reduction of maintenance



The market for inverters for solar installation is quite upbeat following the government's decision to boost solar power generation in the country. Policy action has been supported by corporate backing. With the technology being upgraded over time, solar power inverters have become smarter and offer a number of features that make these more efficient."

— Anurag Garg, vice president, solar BU, Schneider Electric India



cost by countering intermittency in off-grid systems.

In the last decade, lithium-ion based ESSes have helped promote implementation of off-grid solar energy facilities by ensuring a captive source for continuous power supply, therefore minimising dependency on fossil fuels or on-grid systems.

Innovative storage mechanisms (such as flow batteries instead of lead-acid and lithium-ion technologies) and hybridising solar plants with either wind, pumped hydro or biomass can make load balancing more robust.

Powered by smart inverters

Solar system inverters are in great demand following the government's decision to boost solar power generation in the country. Over time, solar power inverters have become smarter and offer a number of features such as remote monitoring, which has increased their demand. Even micro-inverters have received significant importance in the global market. In terms of dimensions, compact inverters in central sizes that have lesser weight and lower losses are growing in popularity.

Solar system developers such as engineering, procurement and construction companies are looking for

inverter manufacturers that offer the latest technology and robust designs, which will enable them to operate in different conditions.

The Indian power industry offers both grid-tie and off-grid inverters. Considering the unstable grids situations, off-grid solar inverters are more suitable as compared to grid-tie inverters. Grid-tie inverters have limited use in India currently, as these require solar as well as grid power.

There is a lot of scope for development in distributed generation of solar hybrid spaces as only a few companies are capable of manufacturing this. Considering the net metering system in India, inverters with both back-up facility and the ability to interact with the grid are expected to be in great demand.

The way forward

Overall, 2015 seems to be a promising year for the Indian solar sector. It could even be considered as the onset of a golden era for solar energy sector if all good intentions of the government turn to reality. However, lack of standardisation for off-grid systems makes designs and performance parameters difficult to achieve. The government needs to establish process guidelines and enforce stringent quality standards to ensure sustainability of off-grid systems.

Last, but not the least, sectorial skill development programmes need to be supported by both the government and industry to ensure availability of huge manpower to support enhanced production and installation of solar systems. ●



More than 50 per cent of the cost for any off-grid solar installation is accounted for by ESSes. Technological innovations are required to develop cheaper and more efficient ESSes."

— Devinder Kumar, chief executive officer, solar projects, Su-Kam

